

// Day 43 of 60

INFRASTRUCTURE

Load Balancing Algorithms

→ Swipe through all 6 slides



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The Traffic Cop Problem

1M users, 5 servers — now what?

Setup:

5 backend servers

1,000,000 requests/sec

Question:

Who decides which server
handles which request?

Answer:

The Load Balancer.



Bad routing = wasted servers



1. Round Robin

The simplest algorithm

```
Request 1 → Server A  
Request 2 → Server B  
Request 3 → Server C  
Request 4 → Server D  
Request 5 → Server E  
Request 6 → Server A
```

...

```
// Works great when all  
// servers are identical
```



Default for most setups



2. Least Connections

Send to the least busy server

```
Server A: 12 active conns  
Server B:  3 active conns ← winner  
Server C:  8 active conns
```

New request → Server B

```
// Perfect for:  
//   long-lived requests  
//   WebSockets, streaming
```



Prevents the slowest server from drowning



3. IP Hashing

Same user → same server

```
hash(client_ip) % num_servers  
→ bucket index
```

```
User John (IP 1.2.3.4)  
→ always Server B
```

```
User Jane (IP 5.6.7.8)  
→ always Server D
```

```
// Useful for sticky sessions  
// and in-memory caches
```



Required when sessions live in RAM



4. Weighted Round Robin

Bigger boxes get more traffic

Server A: 64 cores → weight 8

Server B: 8 cores → weight 1

Out of every 9 requests:

Server A gets 8

Server B gets 1

// Use when running mixed

// hardware in same pool



Match load to actual capacity



Which One to Use?

A quick decision guide

```
Stateless API      → Round Robin
WebSockets         → Least Conns
In-memory session → IP Hash
Mixed hardware     → Weighted RR
```

```
// Tools that support all:
```

```
// • AWS ALB
// • Nginx
// • HAProxy
// • Envoy
```



Picking the right one cuts cloud costs



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